

# CHAPTER 12: STRUCTURE AND PROPERTIES OF CERAMICS

## **ISSUES TO ADDRESS...**

- Structures of ceramic materials:  
How do they differ from that of metals?
- Point defects:  
How are they different from those in metals?
- Impurities:  
How are they accommodated in the lattice and how do they affect properties?
- Mechanical Properties:  
What special provisions/tests are made for ceramic materials?

# CERAMIC BONDING

- Bonding:
  - Mostly ionic, some covalent.
  - % ionic character increases with difference in electronegativity.
- Large vs small ionic bond character:

**CaF<sub>2</sub>: large**  
**SiC: small**

|           |           |                            |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 |               |
|-----------|-----------|----------------------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|---------------|
| IA        |           |                            |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 | 0             |
| H<br>2.1  | IIA       |                            |                 |                 |                 |                 |                 |                 |                 |                 |                 | IIIA            | IVA             | VA              | VIA             | VIIA            | He<br>-       |
| Li<br>1.0 | Be<br>1.5 |                            |                 |                 |                 |                 |                 |                 |                 |                 |                 | B<br>2.0        | C<br>2.5        | N<br>3.0        | O<br>3.5        | F<br>4.0        | Ne<br>-       |
| Na<br>0.9 | Mg<br>1.2 |                            |                 |                 |                 |                 |                 |                 |                 |                 |                 | Al<br>1.5       | Si<br>1.8       | P<br>2.1        | S<br>2.5        | Cl<br>3.0       | Ar<br>-       |
| K<br>0.8  | Ca<br>1.0 | 21<br>Sc<br>1.3            | 22<br>Ti<br>1.5 | 23<br>V<br>1.6  | 24<br>Cr<br>1.6 | 25<br>Mn<br>1.5 | 26<br>Fe<br>1.8 | 27<br>Co<br>1.8 | 28<br>Ni<br>1.8 | 29<br>Cu<br>1.9 | 30<br>Zn<br>1.8 | 31<br>Ga<br>1.6 | 32<br>Ge<br>1.8 | 33<br>As<br>2.0 | 34<br>Se<br>2.4 | 35<br>Br<br>2.8 | 36<br>Kr<br>- |
| Rb<br>0.8 | Sr<br>1.0 | 39<br>Y<br>1.2             | 40<br>Zr<br>1.4 | 41<br>Nb<br>1.6 | 42<br>Mo<br>1.8 | 43<br>Tc<br>1.9 | 44<br>Ru<br>2.2 | 45<br>Rh<br>2.2 | 46<br>Pd<br>2.2 | 47<br>Ag<br>1.9 | 48<br>Cd<br>1.7 | 49<br>In<br>1.7 | 50<br>Sn<br>1.8 | 51<br>Sb<br>1.9 | 52<br>Te<br>2.1 | 53<br>I<br>2.5  | 54<br>Xe<br>- |
| Cs<br>0.7 | Ba<br>0.9 | 57-71<br>La-Lu<br>1.1-1.2  | 72<br>Hf<br>1.3 | 73<br>Ta<br>1.5 | 74<br>W<br>1.7  | 75<br>Re<br>1.9 | 76<br>Os<br>2.2 | 77<br>Ir<br>2.2 | 78<br>Pt<br>2.2 | 79<br>Au<br>2.4 | 80<br>Hg<br>1.9 | 81<br>Tl<br>1.8 | 82<br>Pb<br>1.8 | 83<br>Bi<br>1.9 | 84<br>Po<br>2.0 | 85<br>At<br>2.2 | 86<br>Rn<br>- |
| Fr<br>0.7 | Ra<br>0.9 | 87-103<br>Ac-No<br>1.1-1.7 |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 |               |

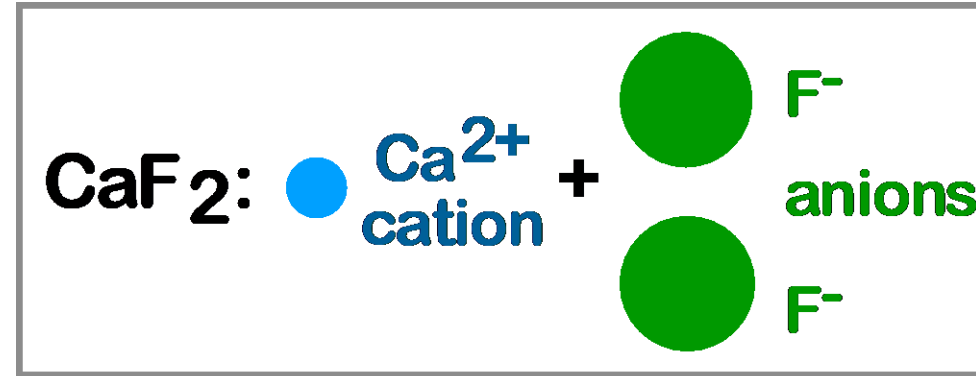
**Table of Electronegativities**

Adapted from Fig. 2.7, *Callister 6e*. (Fig. 2.7 is adapted from Linus Pauling, *The Nature of the Chemical Bond*, 3rd edition, Copyright 1939 and 1940, 3rd edition. Copyright 1960 by Cornell University.

# IONIC BONDING & STRUCTURE

- **Charge Neutrality:**

- Net charge in the structure should be zero.



The anion is larger than cation.

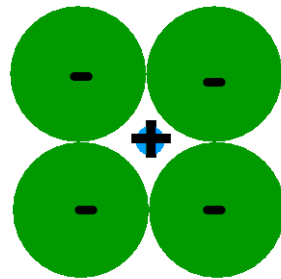
- General form:



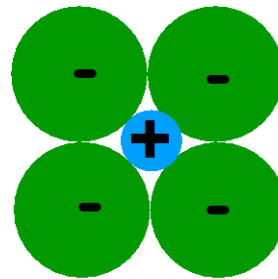
m, p determined by charge neutrality

- **Stable structures:**

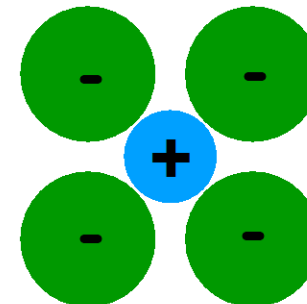
- maximize the # of nearest oppositely charged neighbors.



unstable



stable



stable

Adapted from Fig. 12.1,  
*Callister 6e.*

# COORDINATION # AND IONIC RADII

- Coordination # increases with  $\frac{r_{\text{cation}}}{r_{\text{anion}}}$
- Issue: How many anions can you arrange around a cation?

$\frac{r_{\text{cation}}}{r_{\text{anion}}}$   
Coord #

< .155      2

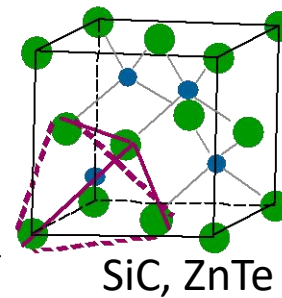
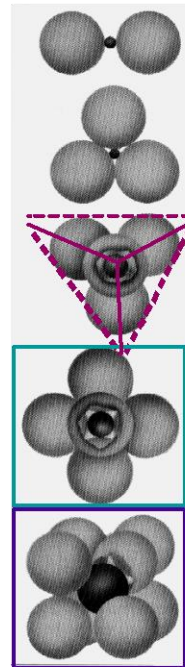
.155-.225      3

.225-.414      4

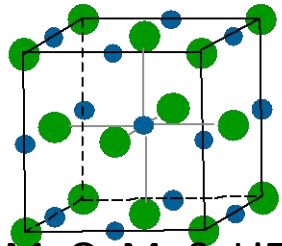
.414-.732      6

.732-1.0      8

Adapted from Table 12.2, Callister 6e.



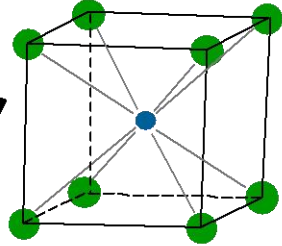
(covalent)  
Zinc Sulfide  
**ZnS**  
(zincblende)  
Adapted from Fig. 12.4, Callister 6e.



**NaCl** (ionic)  
(sodium chloride)

Also MgO, MnS, LiF, and FeO

Adapted from Fig. 12.2, Callister 6e.



**CsCl** (ionic)  
(cesium chloride)  
Adapted from Fig. 12.3, Callister 6e.

Like a BCC structure

### **Computation of Minimum Cation-to-Anion Radius Ratio for a Coordination Number of 3**

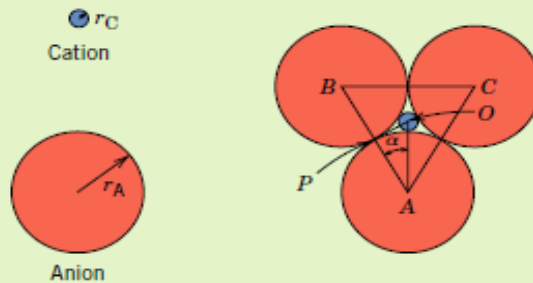
Show that the minimum cation-to-anion radius ratio for the coordination number 3 is 0.155.

### Computation of Minimum Cation-to-Anion Radius Ratio for a Coordination Number of 3

Show that the minimum cation-to-anion radius ratio for the coordination number 3 is 0.155.

#### Solution

For this coordination, the small cation is surrounded by three anions to form an equilateral triangle as shown here, triangle  $ABC$ ; the centers of all four ions are coplanar.



This boils down to a relatively simple plane trigonometry problem. Consideration of the right triangle  $APO$  makes it clear that the side lengths are related to the anion and cation radii  $r_A$  and  $r_C$  as

$$\overline{AP} = r_A$$

and

$$\overline{AO} = r_A + r_C$$

Furthermore, the side length ratio  $\overline{AP}/\overline{AO}$  is a function of the angle  $\alpha$  as

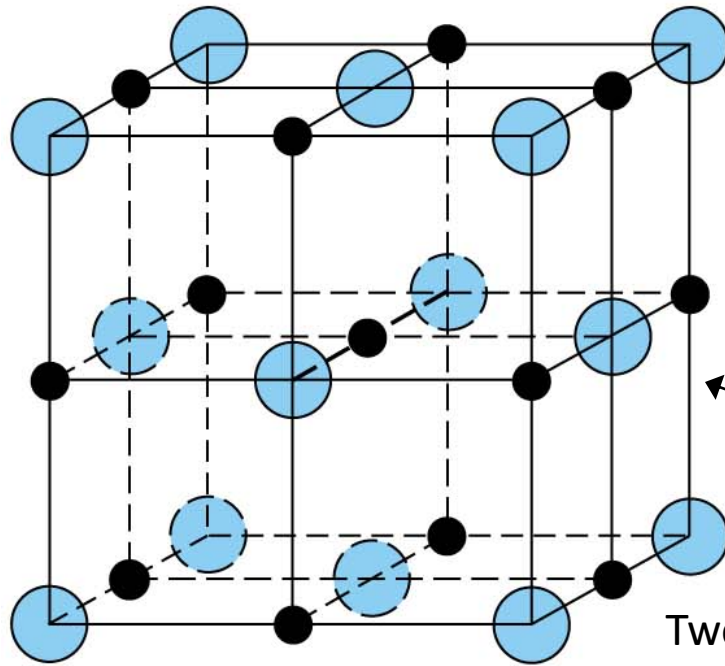
$$\frac{\overline{AP}}{\overline{AO}} = \cos \alpha$$

The magnitude of  $\alpha$  is  $30^\circ$ , because line  $\overline{AO}$  bisects the  $60^\circ$  angle  $BAC$ . Thus,

$$\frac{\overline{AP}}{\overline{AO}} = \frac{r_A}{r_A + r_C} = \cos 30^\circ = \frac{\sqrt{3}}{2}$$

Solving for the cation-anion radius ratio,

$$\frac{r_C}{r_A} = \frac{1 - \sqrt{3}/2}{\sqrt{3}/2} = 0.155$$



**FIGURE 12.2** A unit cell for the rock salt, or sodium chloride (NaCl), crystal structure.

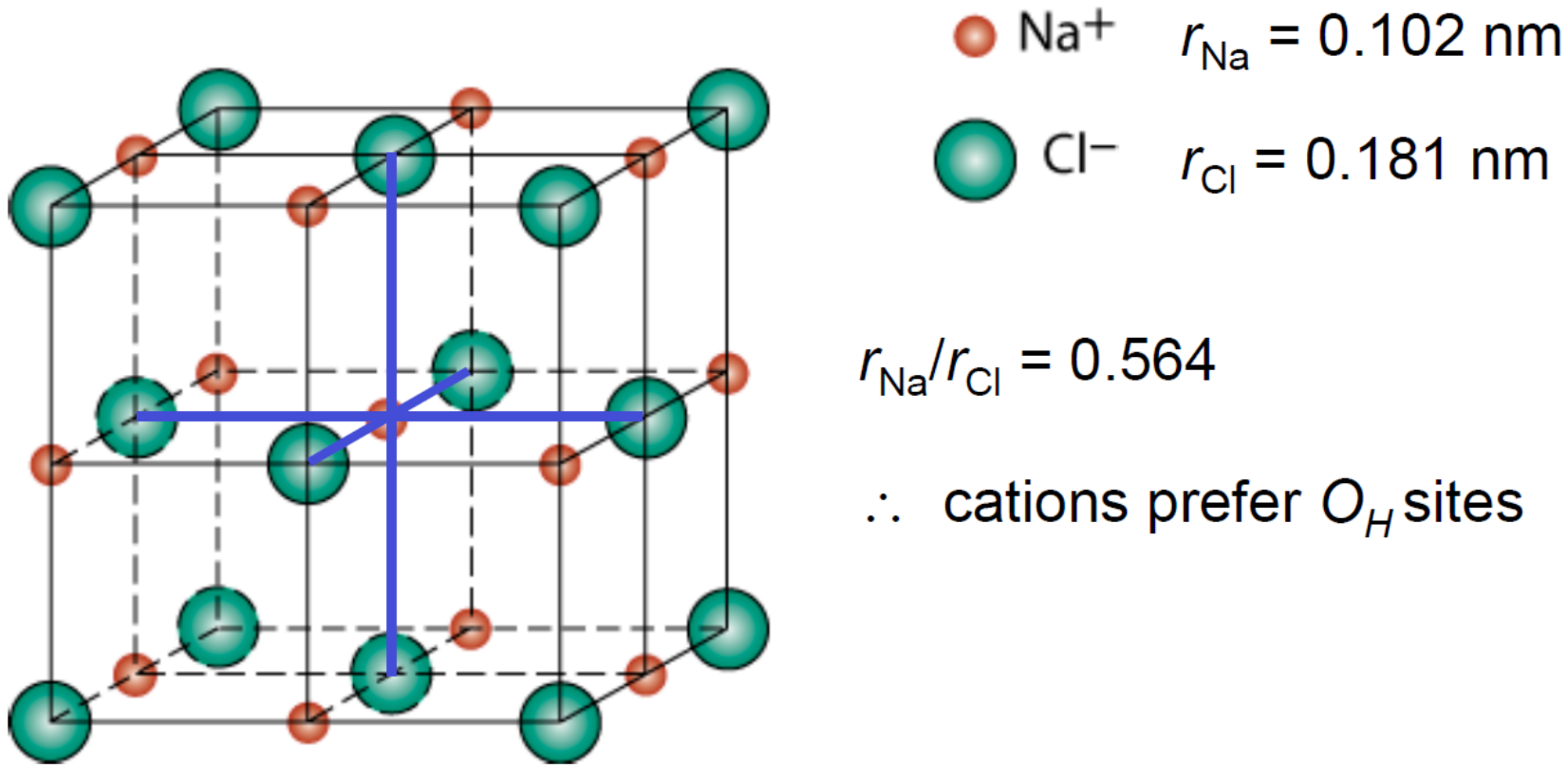
Two interpenetrating FCC lattice,  
one composed of anions and the other of cations

●  $\text{Na}^+$

●  $\text{Cl}^-$

# Rock Salt Structure

Same concepts can be applied to ionic solids in general.  
Example: NaCl (rock salt) structure



Adapted from Fig.  
12.2, *Callister 7e*.

# EX: PREDICTING STRUCTURE OF FeO

- On the basis of ionic radii, what crystal structure would you predict for FeO?

| Cation | Ionic radius (nm) |
|--------|-------------------|
|--------|-------------------|

|                  |       |
|------------------|-------|
| Al <sup>3+</sup> | 0.053 |
|------------------|-------|

|                  |       |
|------------------|-------|
| Fe <sup>2+</sup> | 0.077 |
|------------------|-------|

|                  |       |
|------------------|-------|
| Fe <sup>3+</sup> | 0.069 |
|------------------|-------|

|                  |       |
|------------------|-------|
| Ca <sup>2+</sup> | 0.100 |
|------------------|-------|

Table 12.3 in textbook

| Anion | Ionic radius (nm) |
|-------|-------------------|
|-------|-------------------|

|                 |       |
|-----------------|-------|
| O <sup>2-</sup> | 0.140 |
|-----------------|-------|

|                 |       |
|-----------------|-------|
| Cl <sup>-</sup> | 0.181 |
|-----------------|-------|

|                |       |
|----------------|-------|
| F <sup>-</sup> | 0.133 |
|----------------|-------|

Data from Table 12.3,  
*Callister 6e.*

- Answer:

$$\frac{r_{\text{cation}}}{r_{\text{anion}}} = \frac{0.077}{0.140} = 0.550$$

based on this ratio,  
--coord # = 6  
--structure = NaCl

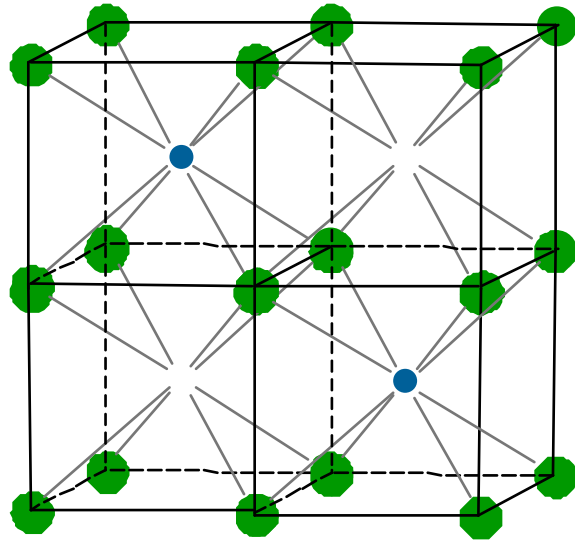
**Table 12.4** Summary of Some Common Ceramic Crystal Structures

| <i>Structure Name</i>       | <i>Structure Type</i>          | <i>Anion Packing</i> | <i>Coordination Numbers</i> |              | <i>Examples</i>                                                     |
|-----------------------------|--------------------------------|----------------------|-----------------------------|--------------|---------------------------------------------------------------------|
|                             |                                |                      | <i>Cation</i>               | <i>Anion</i> |                                                                     |
| Rock salt (sodium chloride) | AX                             | FCC                  | 6                           | 6            | NaCl, MgO, FeO                                                      |
| Cesium chloride             | AX                             | Simple cubic         | 8                           | 8            | CsCl                                                                |
| Zinc blende (sphalerite)    | AX                             | FCC                  | 4                           | 4            | ZnS, SiC                                                            |
| Fluorite                    | AX <sub>2</sub>                | Simple cubic         | 8                           | 4            | CaF <sub>2</sub> , UO <sub>2</sub> , ThO <sub>2</sub>               |
| Perovskite                  | ABX <sub>3</sub>               | FCC                  | 12(A)<br>6(B)               | 6            | BaTiO <sub>3</sub> , SrZrO <sub>3</sub> , SrSnO <sub>3</sub>        |
| Spinel                      | AB <sub>2</sub> X <sub>4</sub> | FCC                  | 4(A)<br>6(B)                | 4            | MgAl <sub>2</sub> O <sub>4</sub> , FeAl <sub>2</sub> O <sub>4</sub> |

**Source:** W. D. Kingery, H. K. Bowen, and D. R. Uhlmann, *Introduction to Ceramics*, 2nd edition. Copyright © 1976 by John Wiley & Sons, New York. Reprinted by permission of John Wiley & Sons, Inc.

# $A_mX_p$ STRUCTURES

- Consider  $\text{CaF}_2$  :  
(Fluorite)  $\frac{r_{\text{cation}}}{r_{\text{anion}}} = \frac{0.100}{0.133} \approx 0.8$
- Based on this ratio, coord # = 8 and structure = CsCl.
- Result: CsCl structure w/only half the **cation** sites occupied.

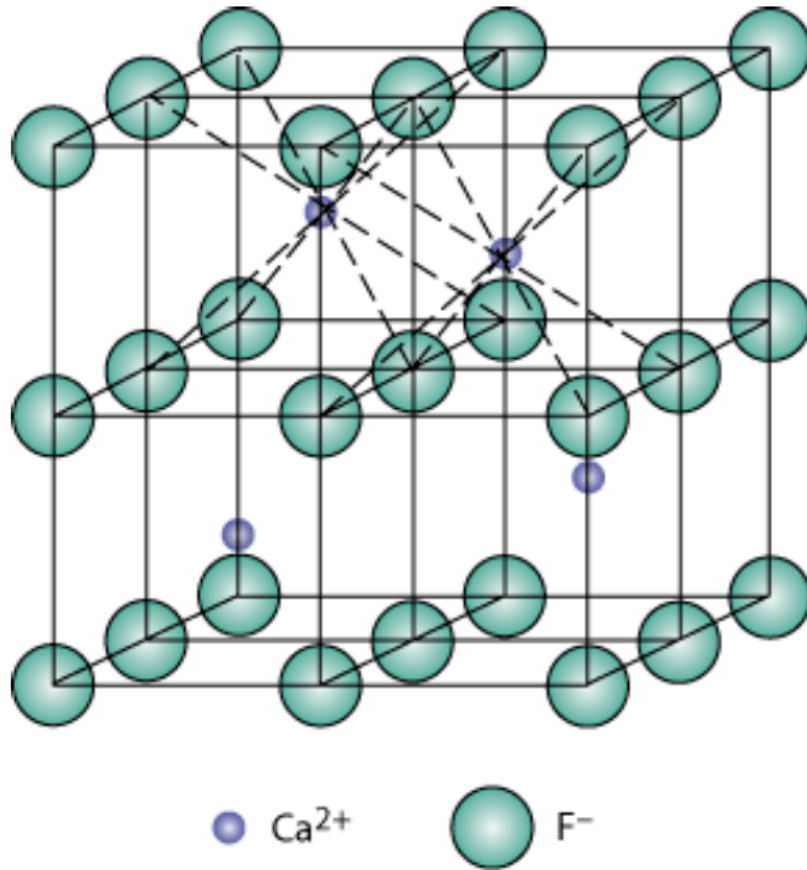


- Only half the **cation** sites are occupied since  $\# \text{Ca}^{2+} \text{ ions} = 1/2 \# \text{F}^- \text{ ions}$ .

Adapted from Fig. 12.5,  
*Callister 6e*.

# AX<sub>2</sub> Crystal Structures

## Fluorite structure



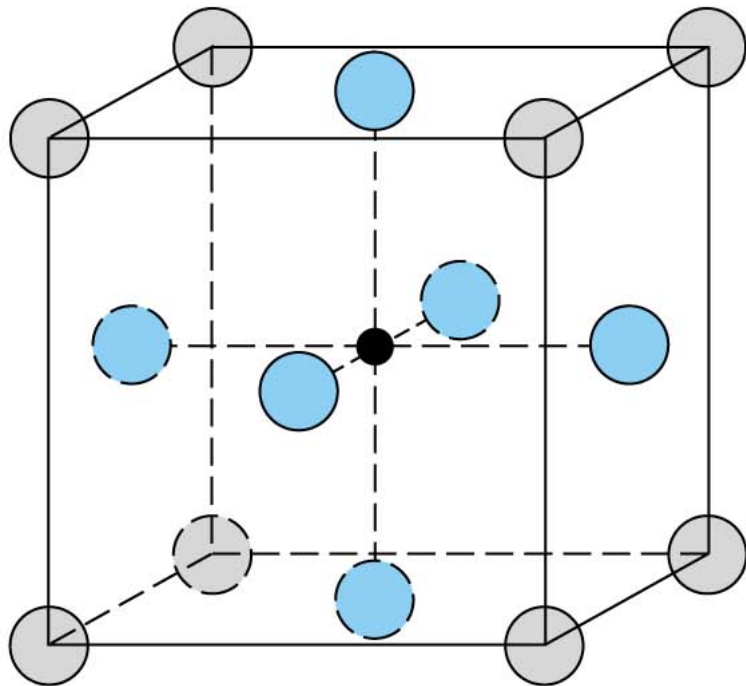
- Calcium Fluorite (CaF<sub>2</sub>)
- cations in cubic sites
- UO<sub>2</sub>, ThO<sub>2</sub>, ZrO<sub>2</sub>, CeO<sub>2</sub>
- antifluorite structure – cations and anions reversed

Adapted from Fig.  
12.5, *Callister 7e*.

Two types of cations

# $A_m B_n X_p$ STRUCTURES

Example: Barium Titanate ( $BaTiO_3$ )



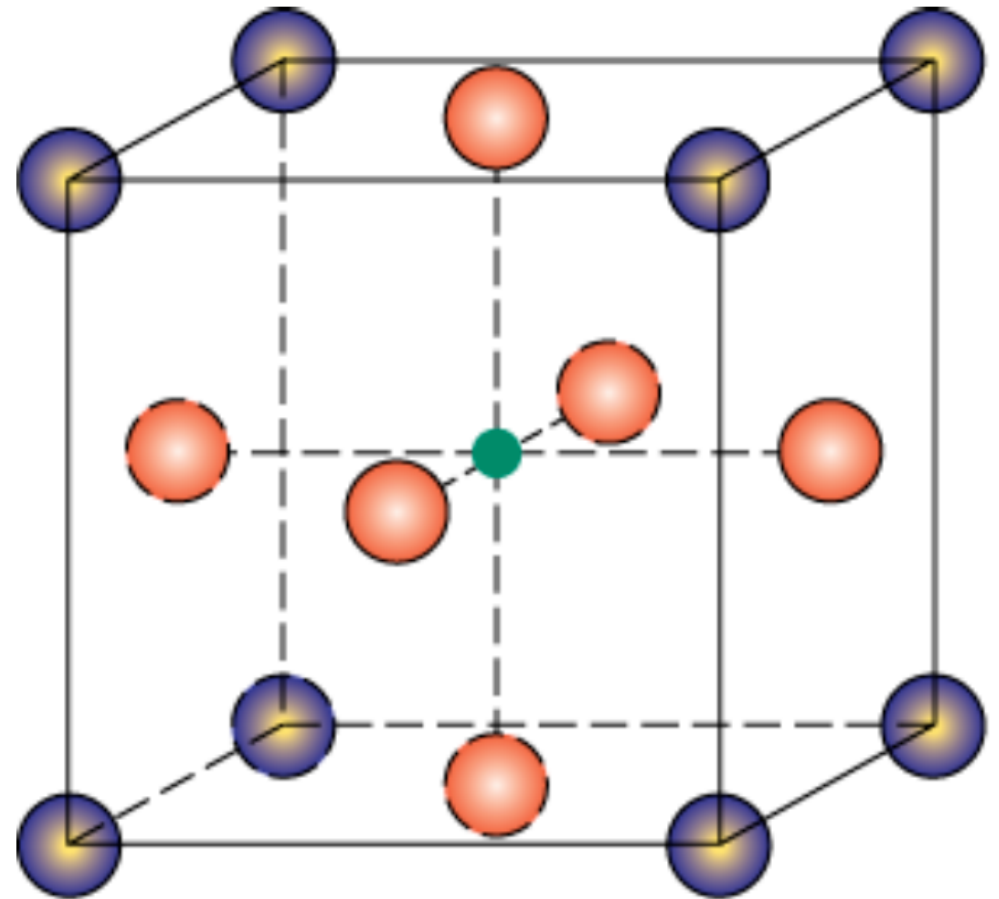
**FIGURE 12.6** A unit cell for the perovskite crystal structure.

# $ABX_3$ Crystal Structures

- Perovskite

Ex: complex oxide  
 $BaTiO_3$

Adapted from Fig.  
12.6, *Callister 7e*.

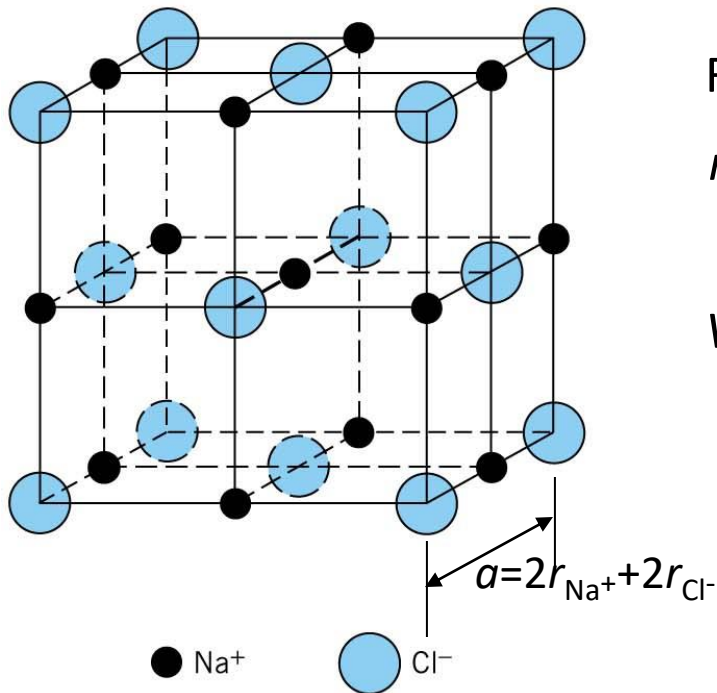


# Ceramic Density Calculations

Number of formula units within the unit cell

$$\rho = \frac{n'(\sum A_C + \sum A_A)}{V_C N_A}$$

Sum of the atomic weights of all anions in the formula unit



Example: density of Sodium Chloride

Formula is NaCl

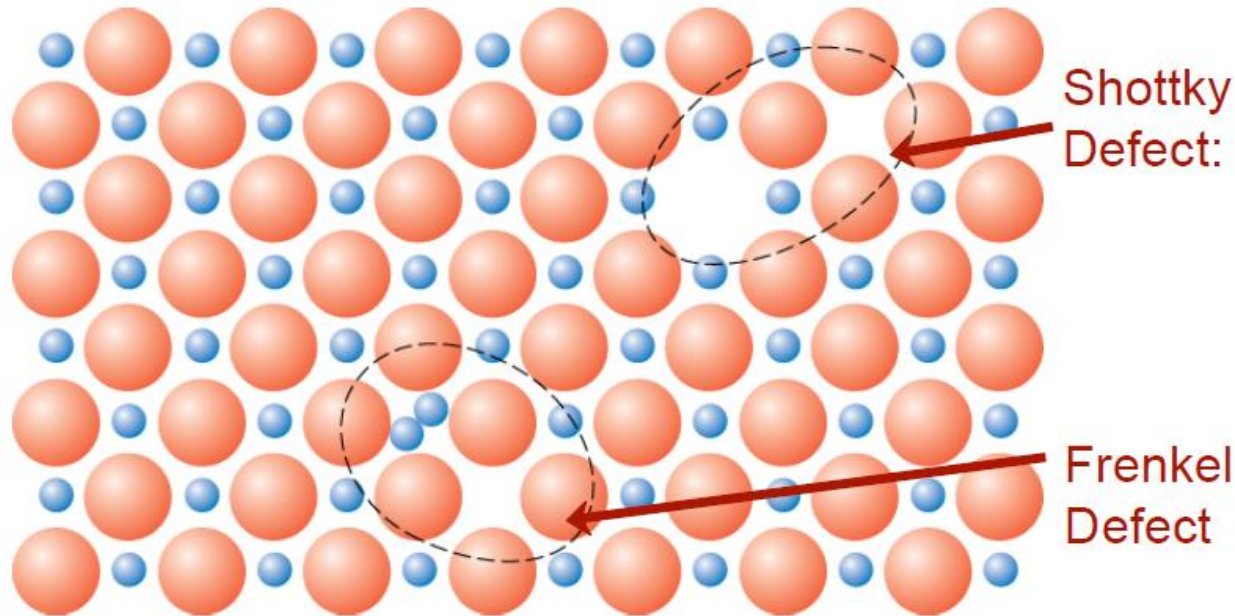
$n' = 4$  (because there are 4 Na atoms and 4 Cl atoms within a unit cell)

$$V_C = a^3 = (2(0.102E-7) + 2(0.181E-7))^3 = (0.566E-7)^3$$

$$\rho = \frac{4(22.99 + 35.45)}{(0.566E-7)^3 (6.023E23)} = 2.14 \text{ g / cm}^3$$

# Defects in Ceramic Structures

- Frenkel Defect  
--a cation is out of place.
- Shottky Defect  
--a paired set of cation and anion vacancies.



Adapted from Fig. 12.21, *Callister* 7e. (Fig. 12.21 is from W.G. Moffatt, G.W. Pearsall, and J. Wulff, *The Structure and Properties of Materials*, Vol. 1, *Structure*, John Wiley and Sons, Inc., p. 78.)

- Equilibrium concentration of defects  $\sim e^{-Q_D / 2kT}$

### Computation of the Number of Schottky Defects in KCl

Calculate the number of Schottky defects per cubic meter in potassium chloride at 500°C. The energy required to form each Schottky defect is 2.6 eV, whereas the density for KCl (at 500°C) is 1.955 g/cm<sup>3</sup>.

#### Solution

To solve this problem it is necessary to use Equation 12.3. However, we must first compute the value of  $N$  (the number of lattice sites per cubic meter); this is possible using a modified form of Equation 4.2:

$$N = \frac{N_A \rho}{A_K + A_{Cl}} \quad (12.4)$$

where  $N_A$  is Avogadro's number ( $6.022 \times 10^{23}$  atoms/mol),  $\rho$  is the density, and  $A_K$  and  $A_{Cl}$  are the atomic weights for potassium and chlorine (i.e., 39.10 and 35.45 g/mol), respectively. Therefore,

$$\begin{aligned} N &= \frac{(6.022 \times 10^{23} \text{ atoms/mol})(1.955 \text{ g/cm}^3)(10^6 \text{ cm}^3/\text{m}^3)}{39.10 \text{ g/mol} + 35.45 \text{ g/mol}} \\ &= 1.58 \times 10^{28} \text{ lattice sites/m}^3 \end{aligned}$$

Now, incorporating this value into Equation 12.3 leads to the following value for  $N_s$ :

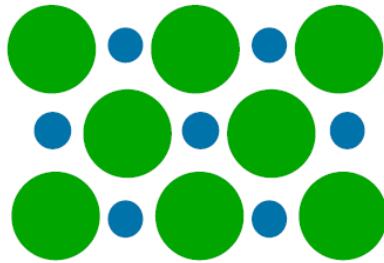
$$\begin{aligned} N_s &= N \exp\left(-\frac{Q_s}{2kT}\right) \\ &= (1.58 \times 10^{28} \text{ lattice sites/m}^3) \exp\left[-\frac{2.6 \text{ eV}}{(2)(8.62 \times 10^{-5} \text{ eV/K})(500 + 273 \text{ K})}\right] \\ &= 5.31 \times 10^{19} \text{ defects/m}^3 \end{aligned}$$

# Impurities

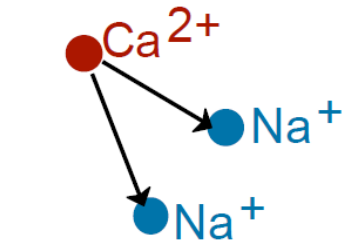
- Impurities must also satisfy **charge balance** = Electroneutrality



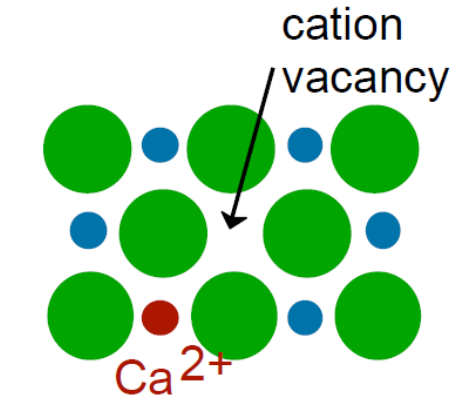
- Substitutional cation impurity



initial geometry

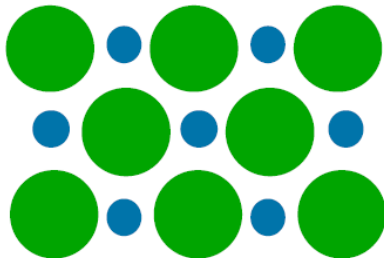


$\text{Ca}^{2+}$  impurity

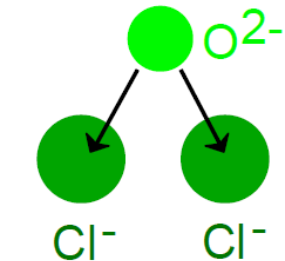


resulting geometry

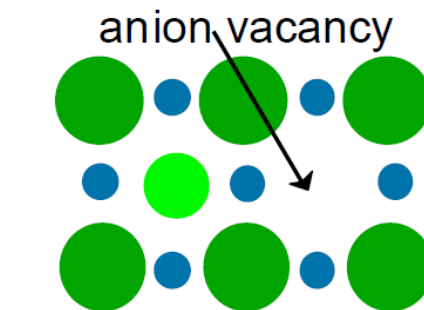
- Substitutional anion impurity



initial geometry



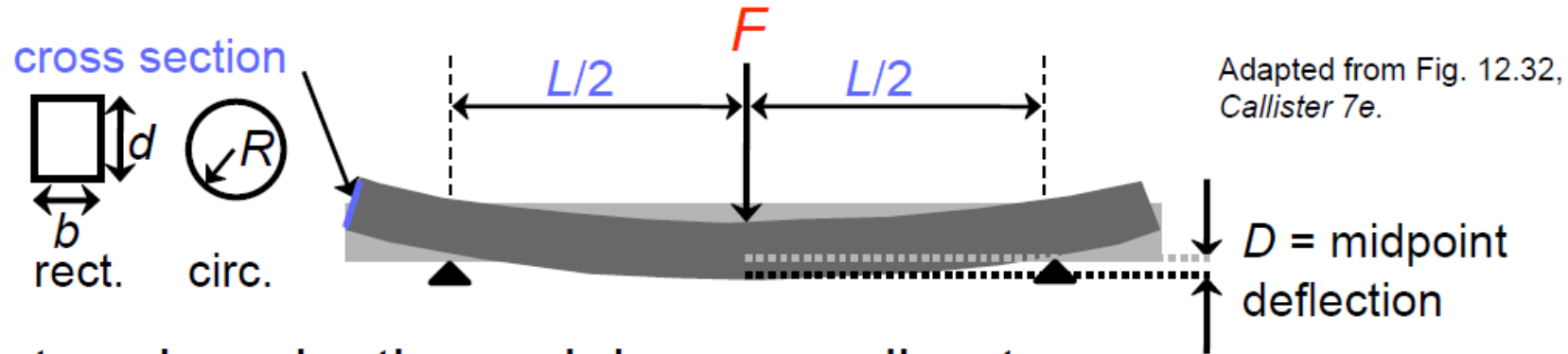
$\text{O}^{2-}$  impurity



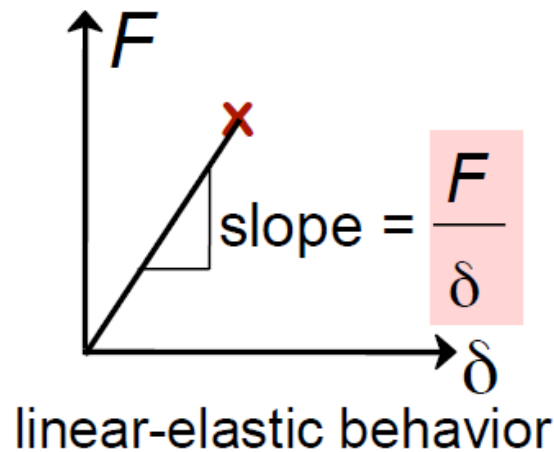
resulting geometry

# Measuring Elastic Modulus

- Room  $T$  behavior is usually elastic, with brittle failure.
- 3-Point Bend Testing often used.
  - tensile tests are difficult for brittle materials.



- Determine elastic modulus according to:



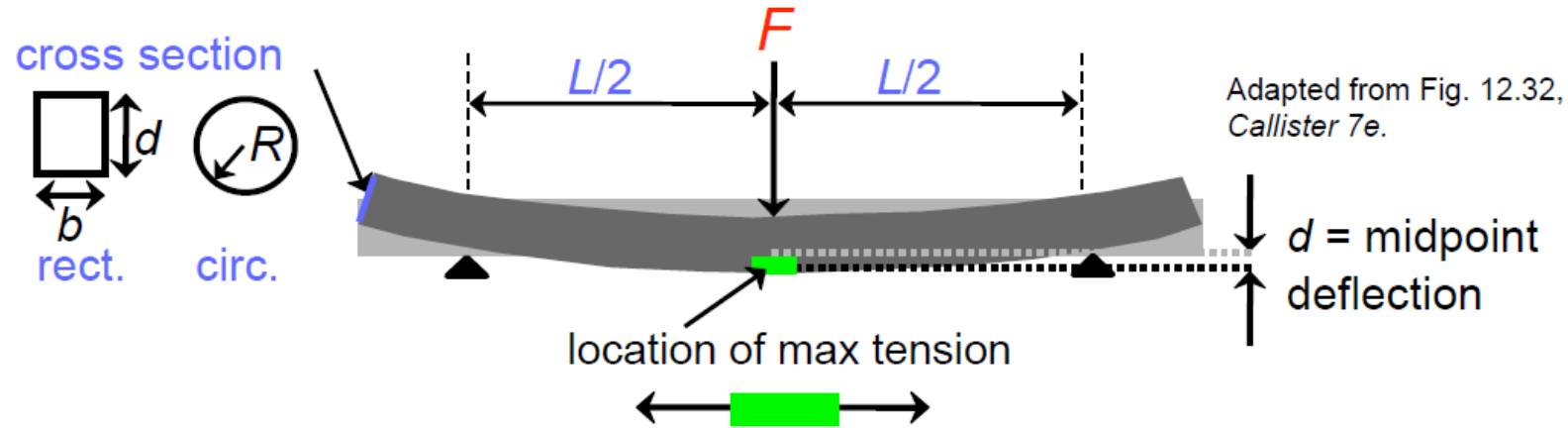
$$E = \frac{F}{\delta} \frac{L^3}{4bd^3} = \frac{F}{\delta} \frac{L^3}{12\pi R^4}$$

rect. cross section      circ. cross section



# Measuring Strength

- 3-point bend test to measure room  $T$  strength.



- Flexural strength:

$$\sigma_{fs} = \frac{1.5F_f L}{bd^2_{\text{rect.}}} = \frac{F_f L}{\pi R^3}$$

$F$

$F_f$

$\delta$

$\delta_{fs}$

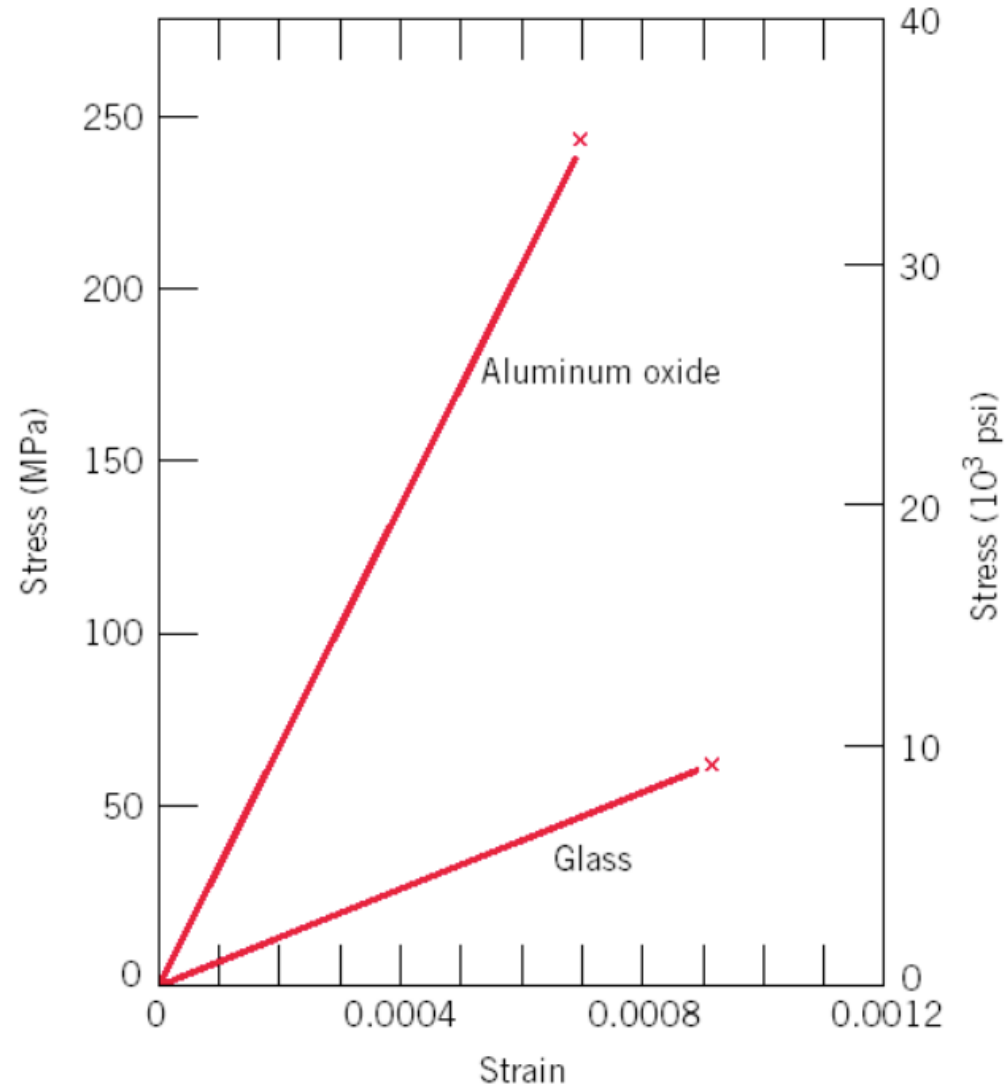
- Typ. values:

| Material     | $\sigma_{fs}$ (MPa) | $E$ (GPa) |
|--------------|---------------------|-----------|
| Si nitride   | 250-1000            | 304       |
| Si carbide   | 100-820             | 345       |
| Al oxide     | 275-700             | 393       |
| glass (soda) | 69                  | 69        |

Data from Table 12.5, Callister 7e.

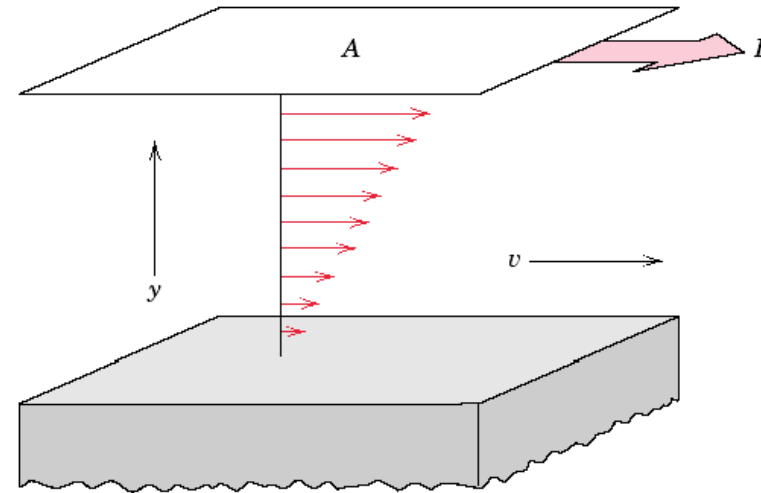
# Elastic behavior

- Typical stress-strain behavior to fracture for aluminum oxide and glass



# Mechanisms of plastic deformation

- **Crystalline ceramics are brittle**
  - Covalent bonds are relatively strong
  - There are limited numbers of slip systems
  - Dislocation structures are complex
- **Noncrystalline ceramics**
  - Plastic deformation does not occur by dislocation motion for noncrystalline ceramics
  - Viscosity is a measure of non-crystalline material's resistance to deformation



# Hardness

**Table 13.6** Approximate Knoop Hardness (100 g load) for Seven Ceramic Materials

| <i>Material</i>              | <i>Approximate Knoop Hardness</i> |
|------------------------------|-----------------------------------|
| Diamond (carbon)             | 7000                              |
| Boron carbide ( $B_4C$ )     | 2800                              |
| Silicon carbide (SiC)        | 2500                              |
| Tungsten carbide (WC)        | 2100                              |
| Aluminum oxide ( $Al_2O_3$ ) | 2100                              |
| Quartz ( $SiO_2$ )           | 800                               |
| Glass                        | 550                               |

# Summary

- Ceramic materials have covalent & ionic bonding.
- Structures are based on:
  - charge neutrality
  - maximizing # of nearest oppositely charged neighbors.
- Structures may be predicted based on:
  - ratio of the cation and anion radii.
- Defects
  - must preserve charge neutrality
  - have a concentration that varies exponentially w/ $T$ .
- Room  $T$  mechanical response is elastic, but fracture is brittle, with negligible deformation.